

Math 548
Portfolio Assignment 7

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Problem 1 (S19, P4). Let R be a commutative ring. For each nonempty subset $X \subseteq R$, the **annihilator** of X is the set $\text{ann}(X) = \{a \in R \mid ax = 0 \text{ for all } x \in X\}$.

- (a) Prove that $\text{ann}(X)$ is an ideal of R .
- (b) Prove that $X \subseteq \text{ann}(\text{ann}(X))$.

Proof. (a) Let $a, b \in \text{ann}(X)$. Then $ax = 0$ and $bx = 0$ for all $x \in X$. Hence $a + b \in \text{ann}(X)$ because $(a + b)x = ax + bx = 0$ for all $x \in X$. Now let $a \in \text{ann}(X)$ and $r \in R$. For any $x \in X$, we have $(ar)x = (ax)r = 0r = 0$, hence $ar \in \text{ann}(X)$. Thus $\text{ann}(X)$ is an ideal of R .

(b) Note that:

$$\text{ann}(\text{ann}(X)) = \{a \in R \mid ax = 0 \text{ for all } x \in \text{ann}(X)\}.$$

Let $y \in X$ and let $b \in \text{ann}(X)$. Then for all $x \in X$, we have $bx = 0$. In particular, $yb = 0$. Since b was arbitrary, $y \in \text{ann}(\text{ann}(X))$. Thus $X \subseteq \text{ann}(\text{ann}(X))$. $\square$

Problem 2 (S24, P3). Let R be a commutative ring with 1, and N the ideal:

$$N = \{a \in R \mid a^n = 0 \text{ for some } n\}.$$

Let $[b]$ be the image of $b \in R$ in R/N . Prove that if $[a] \in R/N$ and $[a]^m = 0$ then $[a] = [0]$.

Proof. Let $[a] \in R/N$. Suppose $[a]^m = [0]$. Then $[a^m] = [0]$, hence $a^m \in N$. But this means $(a^m)^n = 0$ for some n . Hence $a^{mn} = 0$, which means $a \in N$. Since $a \in N$, it must be that $[a] = [0]$. $\square$

Problem 3 (S25, P3). Let $z \in \mathbb{C}$ be a complex number and let $e_z : \mathbb{R}[x] \rightarrow \mathbb{C}$ be the evaluation homomorphism given by $e_z(p) = p(z)$ for each $p \in \mathbb{R}[x]$.

- (a) Show that $\ker(e_z)$ is a prime ideal.
- (b) Compute $\ker(e_{1+i})$, $\text{im}(e_{1+i})$, and then state the conclusion of the First Isomorphism Theorem applied to the homomorphism e_{1+i} .

Proof. (a) By the First Isomorphism Theorem, $\mathbb{R}[x]/\ker(e_z) \cong \text{im}(e_z)$. Since $\mathbb{C}$ is an integral domain, $\text{im}(e_z)$ is an integral domain. Hence $\mathbb{R}[x]/\ker(e_z)$ is an integral domain. Therefore $\ker(e_z)$ is a prime ideal.

(b) Observe that:

$$\begin{aligned} \ker(e_{1+i}) &= \{p \in \mathbb{R}[x] \mid e_{1+i}(p) = 0\} \\ &= \{p \in \mathbb{R}[x] \mid p(1+i) = 0\} \end{aligned}$$

Since $1 + i$ is a root of p , $1 - i$ is also a root. Hence:

$$\begin{aligned} p(x) &= (x - (i - 1))(x - (i + 1)) \\ &= x^2 - 2x + 2. \end{aligned}$$

Since $\mathbf{R}[x]$ is a PID, and since $p(x)$ is an irreducible polynomial contained in the ideal $\ker(\epsilon_{1+i})$, it must be the case that $\ker(\epsilon_{1+i}) = \langle x^2 - 2x + 2 \rangle$. Now consider:

$$\begin{aligned} \text{im}(\epsilon_{1+i}) &= \{ \epsilon_{1+i}(p) \mid p \in \mathbf{R}[x] \} \\ &= \{ p(1 + i) \mid p \in \mathbf{R}[x] \} \end{aligned}$$

Note that the constant polynomials give $\mathbf{R} \subseteq \text{im}(\epsilon_{1+i})$. Moreover, $i \in \text{im}(\epsilon_{1+i})$ since $(x - 1) \in \mathbf{R}[x]$. It must be the case that $\text{im}(\epsilon_{1+i}) = \mathbf{C}$. By the First Isomorphism Theorem, $\mathbf{R}[x]/\langle x^2 - 2x + 2 \rangle \cong \mathbf{C}$. $\square$

Problem 4 (S24, P4). Let $\mathbf{Z}[i] = \{a + bi \mid a, b \in \mathbf{Z}, i^2 = -1\}$, a subring of $\mathbf{C}$. Prove there is no ring homomorphism $\mathbf{Z}[i] \rightarrow \mathbf{Z}_{19}$, but there is a ring homomorphism $\mathbf{Z}[i] \rightarrow \mathbf{Z}_{13}$. Note a ring homomorphism of commutative rings with 1 must send 1 to 1. (Hint: The group of units in $\mathbf{Z}_{19}$ is the cyclic group $\mathbf{U}(19)$ of order 18, and the group of units in $\mathbf{Z}_{13}$ is the cyclic group $\mathbf{U}(13)$ of order 12)

Proof. Suppose towards contradiction there exists a ring homomorphism $\varphi : \mathbf{Z}[i] \rightarrow \mathbf{Z}_{19}$. Note that:

$$\begin{aligned} [1]_{19} &= \varphi(1) \\ &= \varphi(i^4) \\ &= \varphi(i)^4. \end{aligned}$$

Hence $\text{ord}(\varphi(i)) \mid 4$. Case 1: $\text{ord}(\varphi(i)) = 2$. Then:

$$\begin{aligned} [1]_{19} &= \varphi(i)^2 \\ &= \varphi(i^2) \\ &= \varphi(-1) \\ &= -\varphi(1) \\ &= [-1]_{19}. \end{aligned}$$

But this is a contradiction, as the only ring which the multiplicative identity is its own inverse is $\mathbf{Z}/2\mathbf{Z}$. Case 2: $\text{ord}(\varphi(i)) = 1$. If $[1]_{19} = \varphi(i)$, then surely $[1]_{19} = \varphi(i)^2$, which is a contradiction by Case 1. Case 3: $\text{ord}(\varphi(i)) = 4$. Note that $i \in \mathbf{Z}[i]$ is a unit, and since units get mapped to units under ring homomorphisms, it must be the case that $\varphi(i) \in \mathbf{Z}_{19}^*$. But note that $|\mathbf{Z}_{19}^*| = |\mathbf{U}(19)| = 18$. This is a contradiction, as $\text{ord}(\varphi(i)) \nmid 18$. Thus there does not exist a ring homomorphism $\mathbf{Z}[i] \rightarrow \mathbf{Z}_{19}$.

Note that $[5]_{13} \in \mathbf{Z}_{13}$ has an order of 4. Define $\varphi : \mathbf{Z}[i] \rightarrow \mathbf{Z}_{13}$ by $a + ib \mapsto [a + 5b]_{13}$. Observe that:

$$\begin{aligned}\varphi((a + ib) + (c + id)) &= \varphi((a + c) + i(b + d)) \\ &= [(a + c) + 5(b + d)]_{13} \\ &= [(a + 5b) + (c + 5d)]_{13} \\ &= [a + 5b]_{13} + [c + 5d]_{13} \\ &= \varphi(a + ib) + \varphi(c + id).\end{aligned}$$

Similarly:

$$\begin{aligned}\varphi((a + ib)(c + id)) &= \varphi((ac - bd) + i(ad + bc)) \\ &= [(ac - bd) + 5(ad + bc)]_{13} \\ &= [ac + 5ad + 5bc + (-bd)]_{13} \\ &= [ac + 5ad + 5bc + 25bd]_{13} \\ &= [(a + 5b)(c + 5d)]_{13} \\ &= [a + 5b]_{13}[c + 5d]_{13} \\ &= \varphi(a + ib)\varphi(c + id).\end{aligned}$$

Thus φ is a ring homomorphism. □

Problem 5 (F24, P3). Suppose $\varphi : R \rightarrow S$ is a ring homomorphism of commutative rings with 1, and S has no (nonzero) zero-divisors. Prove from the definitions that $\ker(\varphi)$ is a prime ideal.

Proof. Note that $\ker(\varphi)$ is an ideal, so we only need to show that it is prime. Let $x, y \in S$ and $xy \in \ker(\varphi)$. Then:

$$\begin{aligned}0 &= \varphi(xy) \\ &= \varphi(x)\varphi(y).\end{aligned}$$

Since S has no nontrivial zero-divisors, either $\varphi(x) = 0$ or $\varphi(y) = 0$. Hence $x \in \ker(\varphi)$ or $y \in \ker(\varphi)$. Thus $\ker(\varphi)$ is a prime ideal. □